

Final Big RC: Contents Before Mid

University of Michigan-Shanghai Jiaotong University Joint Institute

July 30, 2024

First Physics then Calculus

A person is walking in the night. Light is at point $(L, 0)$ with height H . A person is at point $(l, 0)$, moving to the right with velocity v_0 . Find:

- 1 Velocity of the shadow's head (point S).
- 2 Speed that the length of shadow changes.

Physics and then Calculus-Free Equations

Two objects are placed on a rotating plate, with a string connecting them. The rotating angular velocity of the plate ω is gradually increasing. The mass of the objects are m_1 and m_2 , distances to the axis are R_1 and R_2 . Frictional constant is μ . Find:

- 1 At what ω does tension appears in the string?
- 2 At what ω does the two objects start moving w.r.t on the plate? To which direction?

Considering a Very Small Portion

Consider swinging a test tube filled with liquid with a string (which you probably should not imitate). The two ends of the test tube are at distance x_1 and x_2 from your hand. Area of the tube is S , density of liquid is ρ . Ignore earth gravity. Find:

- 1 Force exerted on the test tube by the string.
- 2 Liquid pressure at distance x from your hand.

